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Short Summary

of

Aeroelastic-Code

ADCoS

*(**A**eroelastic and **D**ynamic **C**omputation **o**f **S**tructures)*

(Version 1.37 / 2.47)

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1. Introduction

The development of wind energy converters (WEC) in the past (in the last years) and the trend to steady increasing turbines requires most exact predicted loads for the structure.
For this, it is necessary, to reproduce fairly well the complete dynamic system (rotor blades, drive train, tower, management) of the WEC in a computer program, for the prediction of the extrem- and fatigue loads under operational conditions.

2. Task

In all standard Finite - Element - Programs (FEM) program modules exist, which calculate the dynamic response of a structure for a given force progress in time of the excitation. Generally this is caused by modal analysis or in a nonlinear FEM program by direct integration. Fact is, that the force progress in time by the aerodynamic forces of a WEC is not known because the forces depend on the movement of the system itself and on the management (power curve of the generator, pitch-control, emergency systems) of the WEC.
Therefor a program system is needed, which considers the coupling of the structural movement and the management on the aerodynamic loads.
In the following the program system *ADCoS Vers. 1.37 / 2.47* is presented, which calculates on the basis of the finite element method the structural answer of a WEC under temporal variable wind loads and under consideration of the management of the WEC.

3. Wind Models

It will be distinguished between a deterministic and a stochastic wind model. As deterministic wind we understand a wind that was taken as known like:

(1) gusts, global on the rotor
all standard forms of gusts
Step gust
Linear gust
Sinus- quadratic
Gusts according to IEC Standard [1]
Positive- and negative gusts

(2) Wind gradient, vertical, horizontal

Definition by Exponent α: $v_h = v_0 \cdot \left(\frac{h}{h_0}\right)^{\alpha}$

Linear above height
Temporal inclined according to IEC Standard [1]
Wind gradient at steady or unsteady atmosphere, the code number is the Obukov- length, see also [9]

(3) Slant streaming, lateral, from below or above
(4) Variation of the streaming because of the tower

The stochastic wind model describes the random variations of the wind speed. In accordance to IEC Standard 61400 – 1 the von Karman turbulence model is used [1]. The longitudinal spectra is given by the non-dimensional equation

$$\frac{f \cdot S_1(f)}{\sigma_1^2} = \frac{4 \cdot f \cdot L / V}{(1 + 71 \cdot (f \cdot L / V)^2)^{5/6}} \qquad (1)$$

where

f frequency in Hz
L is the isotropic length scale, see [1].
σ_1 the longitudinal standard deviation at hub height
V wind velocity at hub height

The lateral and vertical spectra are equal and given by:

$$\frac{f \cdot S_{2,3}(f)}{\sigma_{2,3}^2} = 2 \cdot f \cdot \frac{L}{V} \cdot \frac{1 + 189 \cdot (f \cdot L / V)^2}{(1 + 4 \cdot \pi^2 \cdot (f \cdot L / V)^2)^{1+\gamma}} \qquad (2)$$

where $\sigma_2 = \sigma_3 = \sigma_1$ are the wind speed standard deviation components.

The following exponential coherency model is used

$$Coh(r, f) = \exp\left\{ -8{,}8 \left[(f \cdot r / V)^2 + (0{,}12 \cdot r / L)^2 \right]^{0{,}5} \right\} \qquad (3)$$

where

Coh(r,f) is the coherency function defined by the complex magnitude of the cross-spectral density of the longitudinal wind velocity components at two spatially separated points divided by the autospectrum function.

r is the magnitude of the projection of the separation vector between the two points onto a plane normal to the average direction.

Wind forces on spinner, nacelle and tower are used in both wind models (deterministic and stochastic)

4. Differential Equations of Motion

The general differential equation of motion of an oscillating system in an arbitrary moving coordinate system is given by (including structural damping, centrifugal stiffness, acceleration stiffness, gyroscopic virtual damping in a rotating system)

$$[M]\cdot\ddot{u}+[C+G]\cdot\dot{u}+[K+Z+H]\cdot u=R(t)-R_{I0}(t) \qquad (4)$$

where

$[M]$	Mass matrix
$[C]$	Damping matrix, i. e. rayleigh-damping, see [2]
$[G]$	Coriolis matrix
$[K]$	Stiffness matrix
$[Z]$	Centrifugal stiffness matrix
$[H]$	Acceleration stiffness matrix
$R(t)$	Vector of external loads
$R_{I0}(t)$	Indirect power excitation (only present in rotating systems)
$u,\dot{u},\ddot{u}$	Displacement-, speed- and acceleration vectors

All coefficient matrices have special characteristics. $[M]$, $[C]$, $[K]$ and $[Z]$ are symmetrical, $[G]$ and $[H]$ however are anti-symmetrical. The sum of these matrices are, therefore, generally non-symmetrical. The solution of this system is – as a result of the non - symmetrical coefficient matrices – connected with an enormous input.

Neglecting the coriolis effects, the equation reduces to

$$[M]\cdot\ddot{u}+[C]\cdot\dot{u}+[K+Z]\cdot u=R(t) \qquad (5)$$

(In the aeroelastic code *ADCoS* both equations (eq. (4) and (5)) are implemented)

The equation of motion could be solved in different ways:

(1) Analytical approach, see [7], [8]

(2) Solving the equation by modal analysis, for example *Bladed*. [11]. Normally possible only for the symmetrical problem eq. (5), neglecting non-linear effects.

(3) Non- linear integration in time domain, using finite-lement-method

In the aeroelastic Code *ADCoS* paragraph (3) is used.

Used beam elementes have a cubic displacement interpolation function. Integration is performed using the *von Newmark* integration method, see [2].

In every time step the following equation has to be solved:

$$[K] \cdot x = R \qquad (6)$$

This could be done by a solver like Cholesky [3], [4] or an iterative solver [4], [5] . Note, that for the non- symmetrical problem equation (4) a solver for non-symmetrical matrices must be used.

The finite - element – grid is changed in every new time step depending the new state of the WEC (rotor speed, pitch angle, and so on).

For this, special transformations are needed to transform the displacements, velocities, accelerations of the rotating grid points to a new time step and rotor angle. These transformations must take into account the movement and acceleration of the rotating axis itself and have to be coded avoiding numerical errors.

Advantages of the above described solution are

- Finite- element –method is very flexible, new elements can be added very easy. (for example foundation stiffness, off-shore constructions like tripods or lattice towers)
- The finite-element-grid is adapted in every time step, so a variation of the blade frequency by variation of rotor speed and other coriolis effects are automatically included.
- Non-linear effects like edge- and flap- wise blade- deflection for the pitching moments of the rotor blade are included.
- Dynamic behaviour of the tower torsion is included.

5. Program Description

Input of all necessary data is made using a graphical interface. This is similar to known commercial finite element programs (i. e. *NASTRAN for Windows*)

Input Data

1. General

   air density, rotor diameter, rated power, cone angle, tilt angle, distance rotor-tower

2. Tower

   Tubular Tower: For all tower sections: height, diameter, wall thickness, material, drag coefficient.
   Arbitrary Structure: Element definitions using nodes and element properties

3. Blade

   For all blade sections: radius, chord length, thickness, twist, position of pitch axis, inertia (bending I_Y, I_Z, I_{YZ}, torsion J), material, airfoil data.

4. Nacelle, Yaw-system

   Additional masses, mass inertia, (translational, rotational) area of nacelle fairing and drag coefficients.

5. Stochastic wind model generator

   generalisied von Karman spectra, wind speed, standard deviation, exponent γ, integral length scale, simulation time, wind field size and grid number.

6. Generator

   For example for pitch controlled WEC: momentum- speed- curve

7. Pitch Control

a) internal standard controler
rated speed, time delay, rotor speed for emergency procedure, maximum pitch velocity, normal pitch velocity, minimum output signal, proportional gain, integral gain

b) external (user definded) controler: coded in Lahey FORTRAN LF 95, called from the main program as a dynamic link library (DLL).

8. Material

E-Modulus, Shear-Modulus, densitiy, critical damping, fatigue curve or goodman diagram, safety factors for material

9. Airfoil data

Angle of attack, C_L, C_D, C_M in the range of –180° bis 180°, some parameters for dynamic stall calculation acc. Leishman, Beddoes [10]

10. Loadcases

Calculation time, time step; start up (rotor speed, rotor angle, pitch angle);
wind conditions (gust, wind shear etc.)
Solver flags for calculation.

Coordinate system global:	center tower top	
	X - Axis:	main wind direction
	Z - Axis:	vertical to top
	Y - Axis:	complementary a right-handed coordinate system
Coordinate system element, blade, local:		
	X - Axis:	element direction
	Y - Axis:	global X-direction
	Z - Axis:	complementary a right-handed coordinate system

Coordinate system in rotor plane, rotating and non-rotating

Output data

- For all nodes: time series of displacements in various coordinate systems

- For all elements: node 1 and 2: forces and moments in various coordinate systems (time series)

Postprocessing of time series, like signal analysis, rainflow –counting, fast-fourier-analysis, fatigue calculations, load spectra.

- For a given time and all module units (blades, tower) force- and moment-distribution along blade- or tower- axis in various coordinate systems.

Animation of WEC and blades.

6. Conclusions

With the above described aeroelastic code *ADCoS Vers. 1.37 / 2.47* a powerful tool for calculating the dynamic response of WEC/OWEC is given.

Reaction coupling of aerodynamics, structural movement and WEC management is included. All non-linear behaviour of the rotating, accelerating rotor (virtual gyroscopic damping, angular acceleration stiffness) can be calculated. Torsional stiffness und torsional dynamics including torsional flutter are implemented and validated by other aeroelastic flutter code.

Dynamic torsional response of the tower is included as well. This is very important for lattice towers which have, in fact, a low torsional natural frequency.

The aeroelastic code *ADCoS* has been successfully applied since many years for extrem- and fatigue- load calculations of various WEC up to 5 MW (ENERCON, REpower, Nordex) with different towers (steel, concrete, lattice). Calculations have been validated by „Germanischer Lloyd“, „TÜV Nord“ and „TÜV Süddeutschland“

7. Bibliography Reading List

[1] - IEC 61400 -1 Second Edition 1999-02

[2] Bathe, K.-J. "Finite Element Procedures in Engineering Analysis“ Prentice Hall, Inc, Englewood Cliffs/NJ, USA

[3] Schwarz, H. R. "Methode der Finiten Elemente“ Teubner 1984

[4] Böhn, W. "Methoden der numerischen Mathematik“ Vieweg

[5] Argyris, Mlejnek "Die Methode der Finiten Elemente“, Band II Vieweg

[6] Stoer, Bulisch "Einführung in die numerische Mathematik II“ Springer-Verlag

[7] Schöttl, Ch. "Dynamisches und aeroelastisches Antwortverhalten von Windkraftanlagen“ Dissertation, Universität der Bundeswehr, München, 1991

[8] Hoppe, A. "Modellierung aeroelastischer Effekte an Windkraftanlagen“ Diplomarbeit, Universität Stuttgart, 1995

[9] E. Simiu, R. H. Scanlan "Wind Effects on Structures“ John Wiley & Sons 1996

[10] J. G. Leishman, T. S. Beddoes "A Semi-Empirical Model for Dynamic Stall“ Journal of the American Helicopter Society, 1989

[11] Garrad Hassan and Partners Ltd *„Bladed for Windows“ Vers. 3.393c* A Design Tool for Wind Turbine Performance and Loading

8. Examples

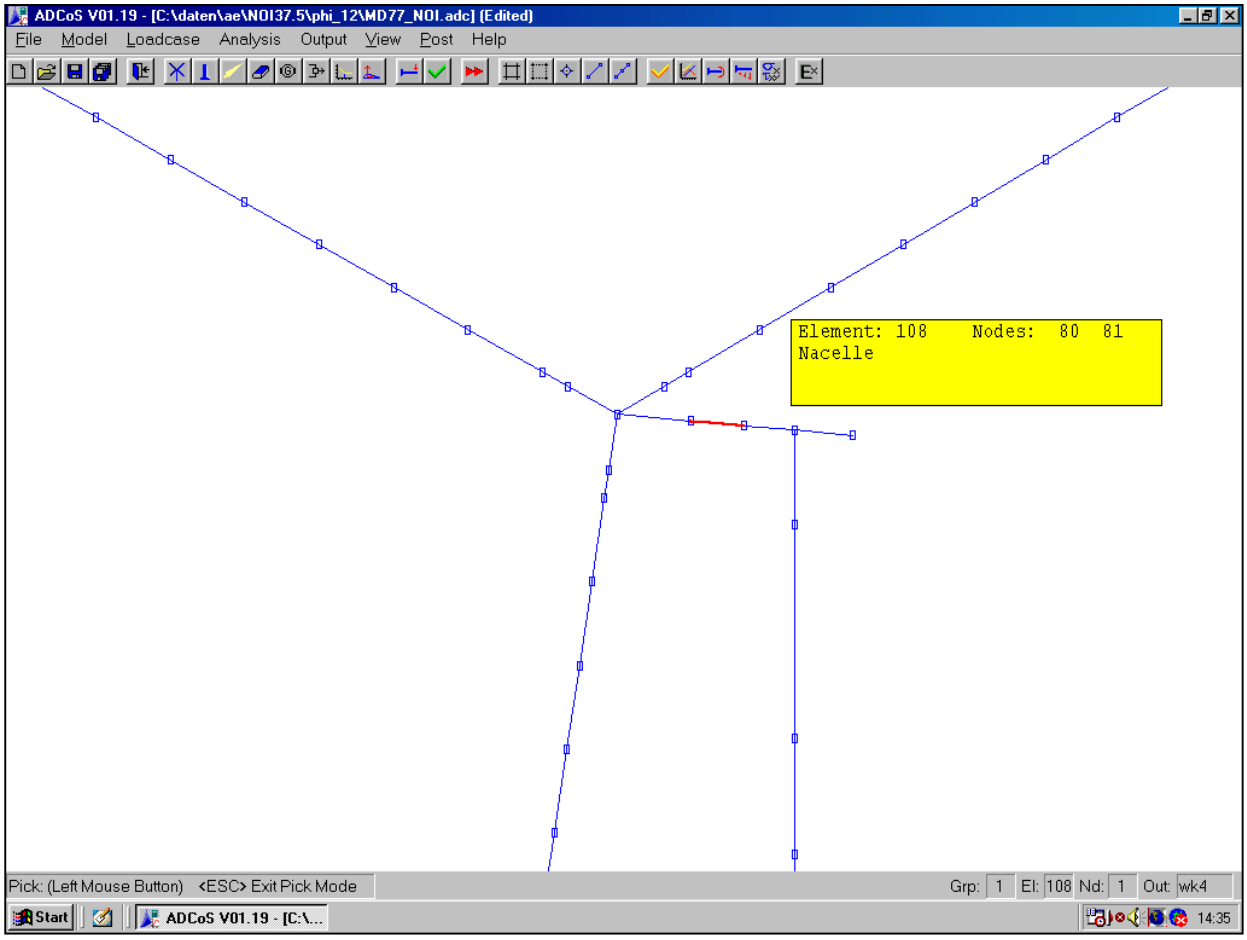

Figure 1: WEC with tubular tower

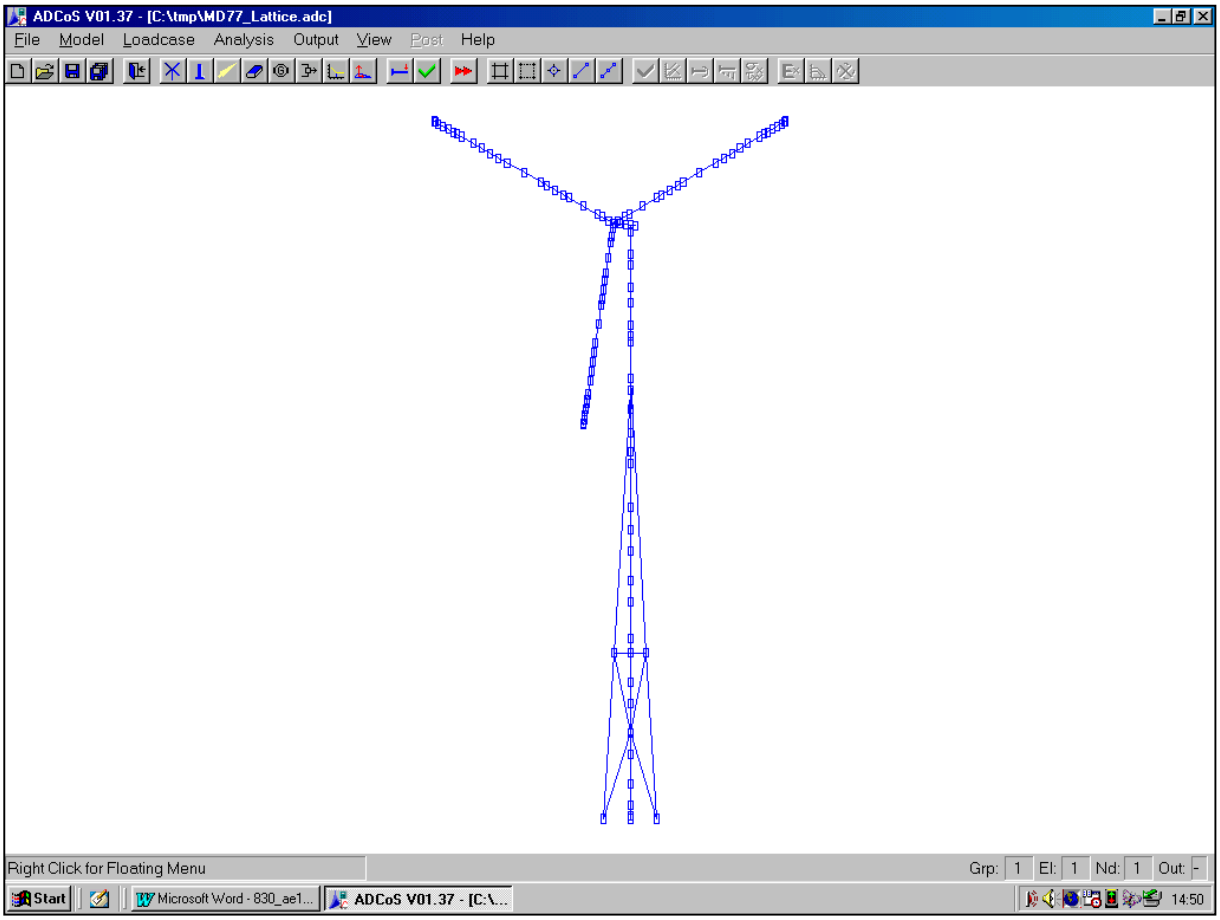

Figure 2: WEC with arbitrary tower structure (like lattice tower)

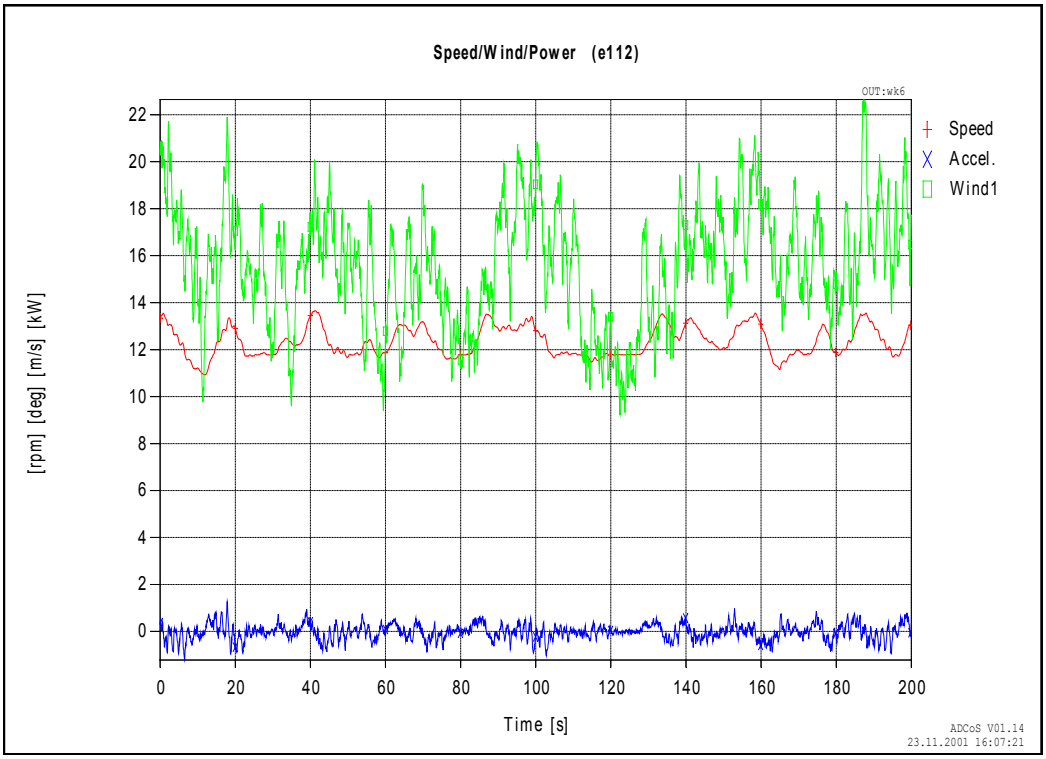

Figure 3: Time series: rotor speed, acceleration, wind.

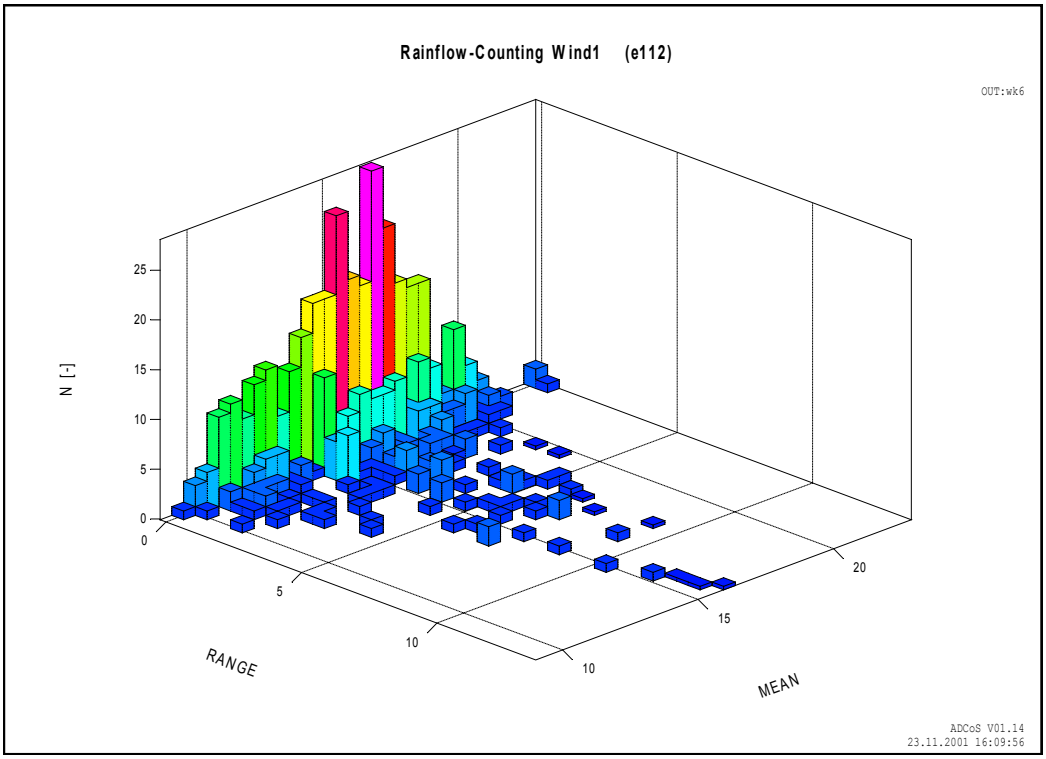

Figure 4: Rainflow- counting

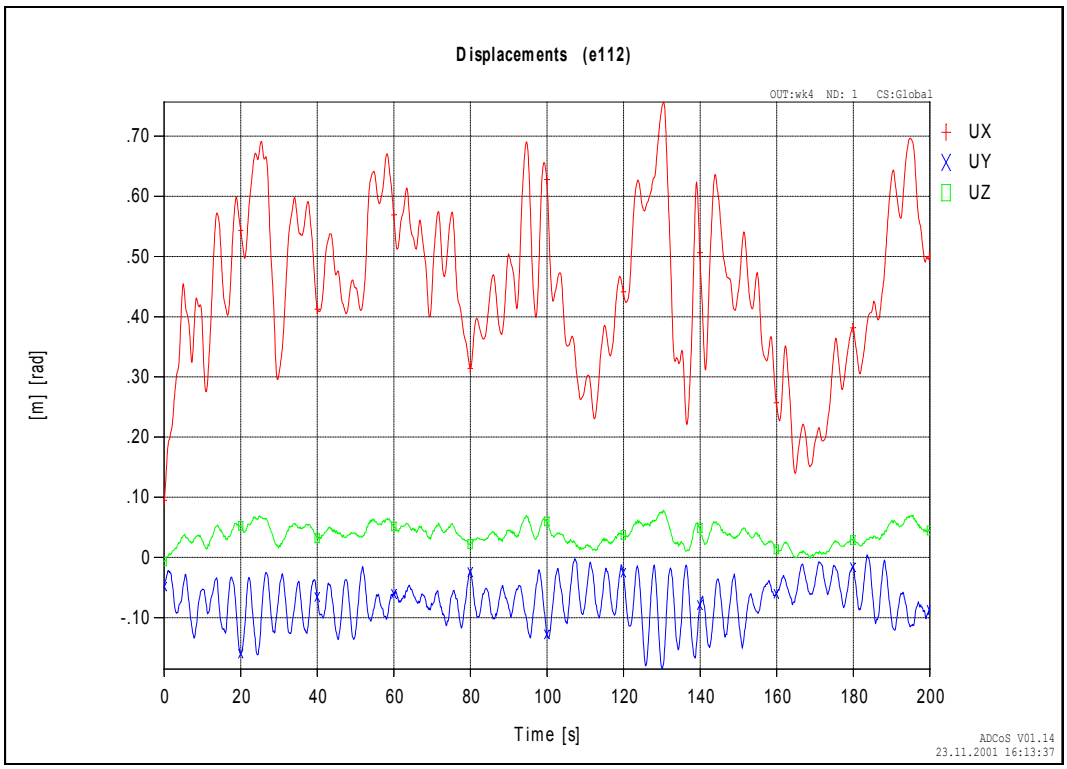

Figure 5: Time series of displacements

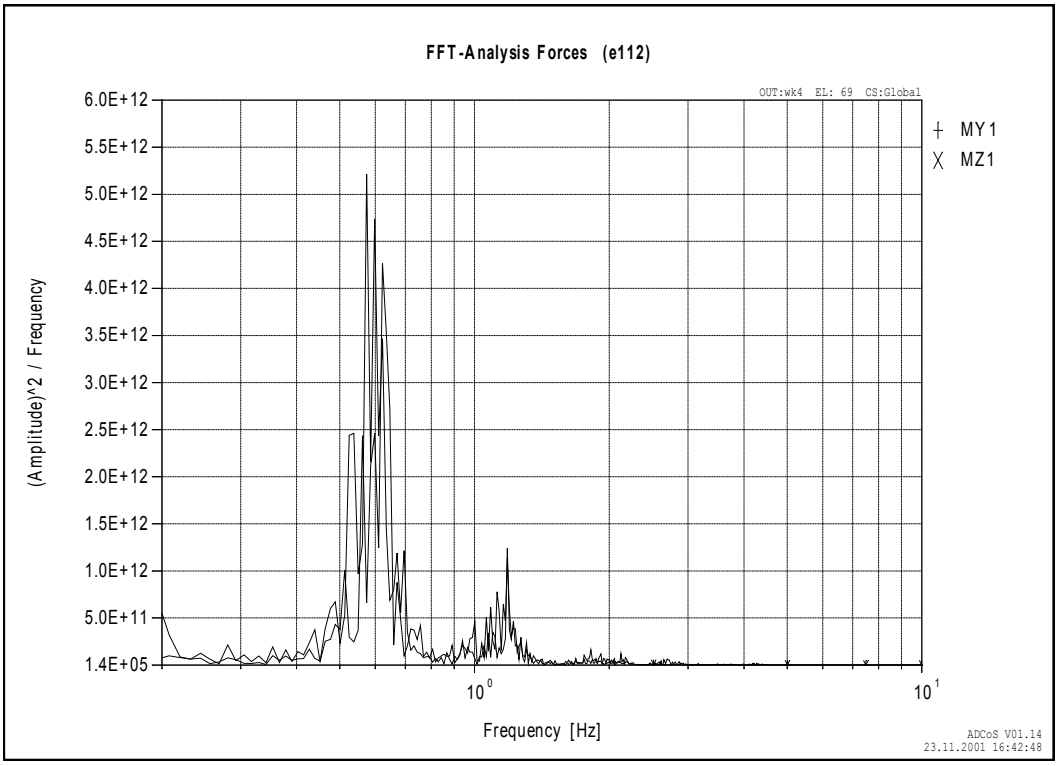

Figure 6: FFT- Analysis

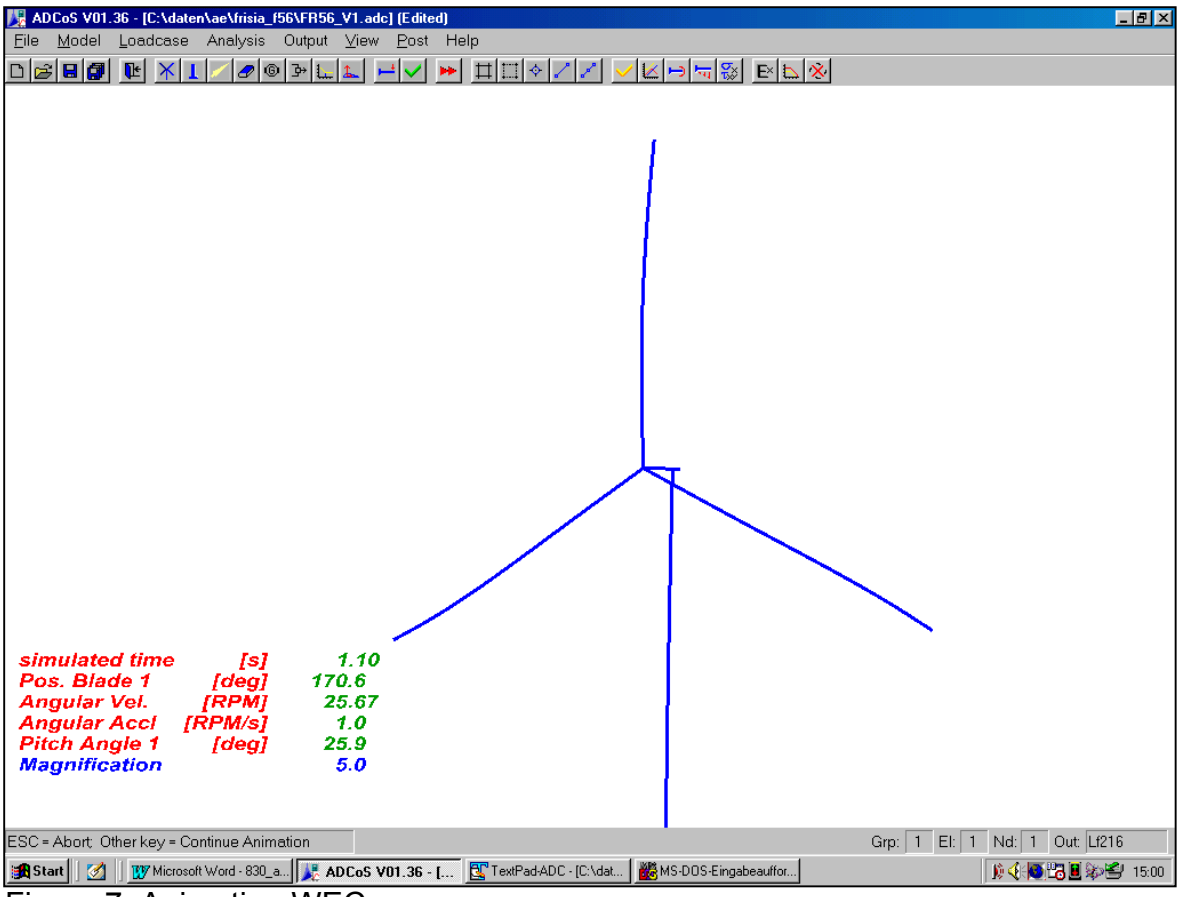

Figure 7: Animation WEC

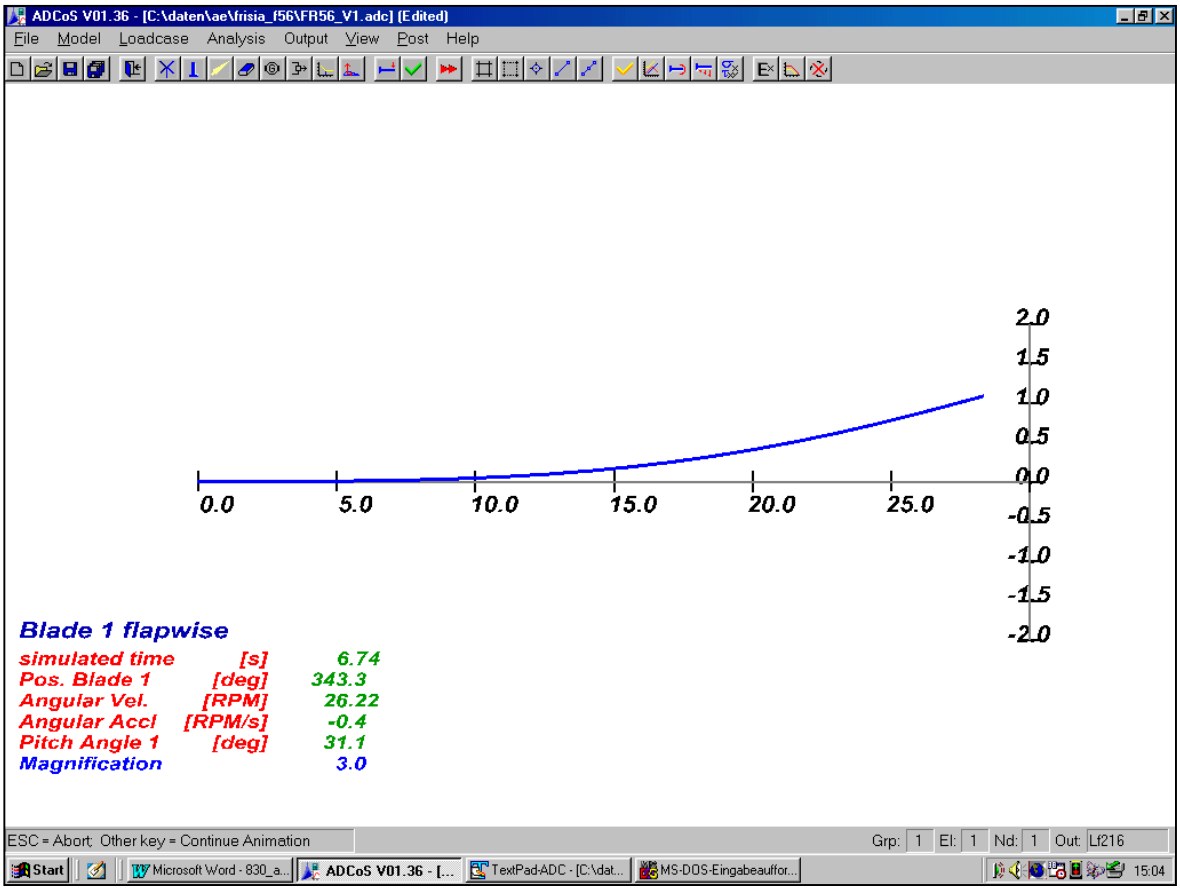

Figure 8: Animation of blade movement: blade 1; flapwise